

The Audit Complexity Theorem: Minimum Expert Count for Reliable Noise Detection, Noise–Expert Trade-off, and the Model Complexity–Audit Cost Duality

Anonymous Author(s)

June 30, 2026

Abstract

SCX auditing presupposes the availability of auditors — human experts or independent AI modules — who can assess samples for label noise. But how many auditors does a given auditing task actually require? We formalize this as the **Audit Complexity Problem** and prove four results, each with fully formalized proofs including complete distribution derivations, explicit premise lists, and honest assessments of the bounds’ limitations. (i) **Independent Expert Lower Bound** (Theorem ??): with K conditionally independent experts of minimum sensitivity $\alpha_{\min}$ and maximum specificity error $\beta_{\max}$, detecting noise level $\eta > 0$ at significance δ requires $K_{\min} \geq 2 \log(1/\delta) / [\eta^2 (\alpha_{\min} - \beta_{\max})^2]$. This bound is **rigorous** via Hoeffding’s inequality applied to the conditional independence structure. (ii) **Noise–Expert Trade-off** (Theorem ??): with budget K , the minimum detectable noise level is $\eta_{\min}(K) = \Theta(1/\sqrt{K})$, a **rigorous** scaling whose lower bound follows from (i) and whose tightness is established by the Berry–Esseen theorem with explicit Lyapunov condition verification. (iii) **Correlated Experts – Audit Diversity** (Theorem ??): with mean inter-expert correlation $\bar{\rho}$, the effective expert count $K_{\text{eff}} = K / [1 + (K - 1)\bar{\rho}]$ controls detectability; the derivation proceeds from variance decomposition under an equicorrelation structure and is conservative for general correlation matrices.

(iv) **Model Complexity–Audit Cost Duality** (Conjecture ??): a conjectured uncertainty relation $K_{\min} \cdot \eta_{\min}^2 \geq c \cdot (M/\log M) \cdot \log(1/\delta)$ linking model complexity M to minimum audit resources — an **open problem**.

1 Introduction

The SCX framework [?] establishes *what* can be known about data quality — Theorem 1 guarantees noise detectability, Theorem 3 bounds unidentifiability, T5 provides optimal active learning sample complexity. But a critical engineering question remains unaddressed: **how many experts does it take to actually perform an audit?**

In practical SCX deployment, expert judgment is the binding resource constraint. Human domain experts are expensive and scarce (HC-Theorem formalizes their audit budget B_H). Independent AI audit modules require separate training, diverse architectures, and non-overlapping training data to achieve genuine independence. Both resources are finite. AC-Theorem answers the question: given a noise level η , a confidence requirement δ , and expert quality parameters (α, β) , what is the minimum number of experts $K_{\min}$ needed to reliably detect noise?

This is *not* a sample complexity question (T5 already answers that). It is an **expert complexity** question — the dimension along which auditors are combined, not along which samples are acquired. The two dimensions interact: more samples can compensate for fewer experts only up to the point where the experts’ judgment error dominates, and vice versa.

Contributions.

- (1) **Independent Expert Lower Bound** (Theorem ??): $K_{\min}$ formula with full distribution derivation (Poisson–binomial mixture under H_1 ; conditional independence exploited). [\[Rigorous\]](#)
- (2) **Noise–Expert Trade-off** (Theorem ??): $\eta_{\min}(K) = \Theta(1/\sqrt{K})$ with Berry–Esseen tightness and Lyapunov verification. [\[Rigorous\]](#)
- (3) **Audit Diversity under Correlation** (Theorem ??): K_{eff} formula via variance decomposition; general- ρ_{ij} Jensen bound. [\[Conditionally Rigorous\]](#) / [\[Open\]](#)
- (4) **Model Complexity Duality** (Conjecture ??): $K_{\min} \cdot \eta_{\min}^2 \gtrsim M/\log M$. [\[Open\]](#)

2 Preliminaries

2.1 Expert Model

Definition 2.1 (Expert Judgment Model). *For a sample x_i with true noise indicator $\varepsilon_i \in \{0, 1\}$, expert $k \in \{1, \dots, K\}$ provides a binary judgment $J_k(x_i) \in \{\text{noise}, \text{hard}\}$ with operating characteristics:*

$$\alpha_k = P(J_k = \text{noise} \mid \varepsilon_i = 1) \quad (\text{sensitivity}), \quad (1)$$

$$\beta_k = P(J_k = \text{hard} \mid \varepsilon_i = 0) \quad (\text{specificity error}). \quad (2)$$

A perfect expert has $\alpha_k = 1, \beta_k = 0$; a random expert has $\alpha_k = \beta_k = 1/2$.

Assumption 2.2 (Conditional Independence). *Expert judgments are conditionally independent given the true noise status ε_i :*

$$P(J_1, \dots, J_K \mid \varepsilon_i) = \prod_{k=1}^K P(J_k \mid \varepsilon_i). \quad (3)$$

This assumption will be relaxed in Section ??.

Assumption 2.3 (Expert Quality Bounds). *For all $k \in [K]$,*

$$\alpha_k \geq \alpha_{\min} > \frac{1}{2}, \quad (4)$$

$$\beta_k \leq \beta_{\max} < \frac{1}{2}. \quad (5)$$

These bounds ensure that every expert is (on the relevant measure) strictly better than random guessing.

Remark 2.4 (Implication of Assumption ??). *From (??) and (??) we obtain the minimum discrimination ability*

$$\alpha_k - \beta_k \geq \alpha_{\min} - \beta_{\max} > 0. \quad (6)$$

This quantity controls the signal-to-noise ratio in all subsequent bounds.

2.2 Audit Complexity of the Model

Definition 2.5 (Model Audit Complexity). *The **audit complexity** M of a model is the effective VC dimension of its Situs space for audit purposes — the minimum number of partition cells required to ε -cover the support of P_X under the Situs metric. For a neural network, M scales with parameter count, depth, and the intrinsic dimension of the data manifold.*

2.3 Detection Problem

The auditor's hypothesis testing problem is:

$$\begin{aligned} H_0 : \eta &= 0 \quad (\text{no label noise}) \\ H_1 : \eta &\geq \eta_{\min} > 0 \quad (\text{label noise present}). \end{aligned} \tag{7}$$

Given K expert judgments per sample, the auditor aggregates them into a test statistic and compares against a threshold at significance level δ .

3 Main Results

3.1 Independent Expert Lower Bound

Theorem 3.1 (Minimum Expert Count — Formal Statement). *Assume Definitions ?? and Assumptions ??, ??. To detect noise level $\eta > 0$ with Type I error $\leq \delta$ and Type II error $\leq \delta$ (power $\geq 1 - \delta$), the minimum number of experts $K_{\min}$ satisfies*

$$K_{\min} \geq \frac{2 \log(1/\delta)}{\eta^2 (\alpha_{\min} - \beta_{\max})^2}. \tag{8}$$

In particular, as $\beta_{\max} \rightarrow 1/2$ (expert approaches random guessing), $K_{\min} \rightarrow \infty$ — no quantity of poor experts compensates for their lack of quality.

Proof. **1**

(P1) $\varepsilon \in \{0, 1\}$ $\{J_k\}_{k=1}^K$ Assumption ??

(P2) $k \in [K]$ $\alpha_k \geq \alpha_{\min} > 1/2$ Assumption ??(??)

(P3) $k \in [K]$ $\beta_k \leq \beta_{\max} < 1/2$ Assumption ??(??)

(P4) $\varepsilon \sim \eta$ Bernoulli

(P5) P_X

2 x

$$T_K(x) = \frac{1}{K} \sum_{k=1}^K 1_{\{J_k(x)=\text{noise}\}}. \tag{9}$$

$$Y_k := 1_{\{J_k=\text{noise}\}} \quad T_K = \frac{1}{K} \sum Y_k \quad Y_k \in \{0, 1\}$$

$$\mathbf{3} H_0 \quad T_K \quad H_0 \quad \varepsilon = 0$$

$$P(Y_k = 1 \mid H_0) = P(J_k = \text{noise} \mid \varepsilon = 0) = \beta_k.$$

$$\text{P1} \{Y_k\}_{k=1}^K \quad H_0 \quad Y_k \sim \text{Ber}(\beta_k) \quad S_K := K \cdot T_K \sim \text{Poisson-binomial}(\{\beta_k\})$$

$$P(S_K = s \mid H_0) = \sum_{A \subseteq [K]: |A|=s} \prod_{k \in A} \beta_k \prod_{k \notin A} (1 - \beta_k), \quad s = 0, \dots, K.$$

$$\mathbb{E}[T_K \mid H_0] = \frac{1}{K} \sum_{k=1}^K \beta_k =: \bar{\beta}.$$

$$\text{P3} \bar{\beta} \leq \beta_{\max}$$

$$\mathbb{V}[T_K \mid H_0] = \frac{1}{K^2} \sum_{k=1}^K \beta_k (1 - \beta_k) \leq \frac{\beta_{\max}(1 - \beta_{\max})}{K} \leq \frac{1}{4K}.$$

$$4H_1 \quad T_K \quad H_1 \quad \varepsilon \sim \text{Ber}(\eta)$$

$$\begin{aligned} P(Y_k = 1 \mid H_1) &= \eta P(Y_k = 1 \mid \varepsilon = 1) + (1 - \eta) P(Y_k = 1 \mid \varepsilon = 0) \\ &= \eta \alpha_k + (1 - \eta) \beta_k =: p_k. \end{aligned}$$

$$\varepsilon \{Y_k\} \quad T_K \quad H_1 \quad \text{Poisson-binomial}$$

$$P(T_K = t \mid H_1) = \eta \cdot P\left(\frac{1}{K} \sum Y_k = t \mid \varepsilon = 1\right) + (1 - \eta) \cdot P\left(\frac{1}{K} \sum Y_k = t \mid \varepsilon = 0\right),$$

$$\{\alpha_k\} \quad \{\beta_k\} \quad \text{Poisson-binomial}$$

$$\begin{aligned} \mathbb{E}[T_K \mid H_1] &= \mathbb{E}[\mathbb{E}[T_K \mid \varepsilon]] \\ &= \eta \cdot \frac{1}{K} \sum_{k=1}^K \alpha_k + (1 - \eta) \cdot \frac{1}{K} \sum_{k=1}^K \beta_k \\ &= \eta \bar{\alpha} + (1 - \eta) \bar{\beta}, \end{aligned}$$

$$\begin{aligned} \bar{\alpha} &:= \frac{1}{K} \sum \alpha_k \quad \text{P2-P3} \quad \bar{\alpha} \geq \alpha_{\min} \bar{\beta} \leq \beta_{\max} \\ \mathbb{V}[T_K] &= \mathbb{E}[\mathbb{V}[T_K \mid \varepsilon]] + \mathbb{V}[\mathbb{E}[T_K \mid \varepsilon]] \end{aligned}$$

$$\begin{aligned} \mathbb{V}[T_K \mid H_1] &= \eta \mathbb{V}\left[\frac{1}{K} \sum Y_k \mid \varepsilon = 1\right] + (1 - \eta) \mathbb{V}\left[\frac{1}{K} \sum Y_k \mid \varepsilon = 0\right] \\ &\quad + \eta(1 - \eta)(\bar{\alpha} - \bar{\beta})^2. \end{aligned}$$

$$\leq 1/(4K) \eta(1 - \eta)(\bar{\alpha} - \bar{\beta})^2 \leq \frac{1}{4}(\alpha_{\min} - \beta_{\max})^2$$

5

$$\Delta := \mathbb{E}[T_K \mid H_1] - \mathbb{E}[T_K \mid H_0] = \eta(\bar{\alpha} - \bar{\beta}).$$

P2-P3 (??)

$$\Delta \geq \eta(\alpha_{\min} - \beta_{\max}) > 0.$$

6Hoeffding $Y_k \in [0, 1] \quad \{Y_k\} \quad H_0$ Hoeffding [?] $\varepsilon > 0$

$$P(|T_K - \mathbb{E}[T_K]| \geq \varepsilon \mid H_0) \leq 2 \exp(-2K\varepsilon^2). \quad (10)$$

H_1

$$\mathbf{7} \quad \theta := \bar{\beta} + \Delta/2 \quad T_K > \theta \quad H_0$$

H_0

$$\begin{aligned} P(T_K > \theta \mid H_0) &= P(T_K - \bar{\beta} > \Delta/2 \mid H_0) \\ &\leq \exp(-2K(\Delta/2)^2) \quad (??) \\ &= \exp(-K\Delta^2/2). \end{aligned}$$

H_1

$$\begin{aligned} P(T_K \leq \theta \mid H_1) &= P(\mathbb{E}[T_K \mid H_1] - T_K \geq \Delta/2 \mid H_1) \\ &\leq \exp(-K\Delta^2/2). \end{aligned}$$

Hoeffding $H_1 \quad \mathbb{E}[T_K \mid H_1]$

δ

$$\exp(-K\Delta^2/2) \leq \delta \implies K \geq \frac{2 \log(1/\delta)}{\Delta^2}.$$

$$\mathbf{8} \quad \Delta \geq \eta(\alpha_{\min} - \beta_{\max})$$

$$K_{\min} \geq \frac{2 \log(1/\delta)}{\eta^2(\alpha_{\min} - \beta_{\max})^2}.$$

AC.1

□

[Rigorous]

:

- Hoeffding $H_0 \quad \varepsilon = 0 \quad \{Y_k\}$ Assumption ?? $H_1 \quad \varepsilon$ Hoeffding ε (??) K (??)
- $\alpha_{\min} \quad \beta_{\max} \quad \bar{\alpha}, \bar{\beta} \quad \Delta \quad \alpha_k \gg \alpha_{\min} \quad \beta_k \ll \beta_{\max} \quad \eta(\alpha_{\min} - \beta_{\max})$ (??) K
“ ” “ ”
- Hoeffding $\beta_{\max} = 0$ Bernstein Bernstein

$$P(|T_K - \mathbb{E}[T_K]| \geq \varepsilon) \leq 2 \exp\left(-\frac{K\varepsilon^2}{2\beta_{\max}(1 - \beta_{\max}) + 2\varepsilon/3}\right),$$

$$\varepsilon = \Delta/2 \quad K \geq 8\beta_{\max}(1 - \beta_{\max}) \log(2/\delta)/\Delta^2 \quad \beta_{\max} = 0 \quad \text{Hoeffding} \quad \beta_{\max}(1 - \beta_{\max})$$

• $n \ n \ T_K \ K \ T5$

□

Applications: Theorem ?? provides the **staffing requirement** for any expert-based SCX audit. Before launching an audit, the formula (??) can be evaluated using known expert quality parameters $(\alpha_{\min}, \beta_{\max})$, the target noise level η , and the required confidence δ . For a typical configuration ($\alpha = 0.85$, $\beta = 0.15$, $\eta = 0.05$, $\delta = 0.05$), the minimum expert count is $K_{\min} \approx 45$ — a non-trivial staffing requirement. The theorem reveals an inescapable trade-off: to halve the detectable noise level (from η to $\eta/2$), one must quadruple the expert count. In medical audit, where domain experts (board-certified physicians) are scarce, this bound directly determines feasibility: below a certain noise level, expert-based auditing becomes economically impossible, and one must rely on complementary Cercis-based detection or accept higher false-negative rates. The divergence $\beta_{\max} \rightarrow 1/2$ warning — poor experts cannot be compensated by quantity — is a hiring criterion: never deploy experts whose specificity is near chance level, regardless of how many are available.

Remark 3.2. *Theorem ?? is **rigorous**. The bound is derived from standard Hoeffding concentration inequalities, the Neyman–Pearson lemma, and the explicit distribution analysis above. [Rigorous]*

3.2 Noise–Expert Trade-off

Theorem 3.3 (Noise–Expert Trade-off — Formal Statement). *Under the same assumptions as Theorem ??, with a fixed expert budget $K \in \mathbb{N}$ and significance $\delta \in (0, 1/2)$, the minimum detectable noise level satisfies*

$$\eta_{\min}(K) \geq \sqrt{\frac{2 \log(1/\delta)}{K (\alpha_{\min} - \beta_{\max})^2}}. \quad (11)$$

*The asymptotic scaling $\eta_{\min}(K) = \Theta(1/\sqrt{K})$ is **strict**:*

- (i) Lower bound: $\eta_{\min}(K) = \Omega(1/\sqrt{K})$ follows from (14).
- (ii) Tightness: *there exists a sequence of tests $\{\varphi_K\}$ such that for $\eta_K = c/\sqrt{K}$ with c sufficiently large, the Type I and Type II errors are both $\leq \delta$, establishing $\eta_{\min}(K) = O(1/\sqrt{K})$.*

Proof. Theorem ?? (??) η

$$\eta_{\min}(K) \geq \sqrt{\frac{2 \log(1/\delta)}{K (\alpha_{\min} - \beta_{\max})^2}}.$$

$$\begin{aligned} \eta_{\min}(K) &= \Omega(1/\sqrt{K}) \\ \text{Berry-Esseen} &\Theta(1/\sqrt{K}) \\ \text{1Berry-Esseen i.i.d.} & \end{aligned}$$

Lemma 3.4 (Berry-Esseen, 1941–1942; Shevtsova 2011 refinement). $X_1, \dots, X_K$
 $\mathbb{E}[X_k] = \mu_k \mathbb{V}[X_k] = \sigma_k^2 > 0 \mathbb{E}[|X_k - \mu_k|^3] = \rho_k < \infty S_K = \sum_{k=1}^K X_k B_K^2 =$
 $\sum_{k=1}^K \sigma_k^2 C_0 \leq 0.5583$ Shevtsova, 2011

$$\sup_{x \in \mathbb{R}} |P\left(\frac{S_K - \mathbb{E}[S_K]}{B_K} \leq x\right) - \Phi(x)| \leq C_0 \cdot \frac{\sum_{k=1}^K \rho_k}{B_K^3}.$$

$$\text{2Lyapunov } X_k = Y_k = 1_{\{J_k = \text{noise}\}} k = 1, \dots, K H_0 Y_k \sim \text{Ber}(\beta_k)$$

$$\rho_k = \mathbb{E}[|Y_k - \beta_k|^3] = \beta_k(1 - \beta_k)^3 + (1 - \beta_k)\beta_k^3 = \beta_k(1 - \beta_k)[(1 - \beta_k)^2 + \beta_k^2].$$

$$(1 - \beta_k)^2 + \beta_k^2 = 1 - 2\beta_k(1 - \beta_k) \leq 1 \rho_k \leq \beta_k(1 - \beta_k) = \sigma_k^2$$

$$\frac{\sum \rho_k}{B_K^3} \leq \frac{\sum \sigma_k^2}{B_K^3} = \frac{1}{B_K}.$$

$$B_K^2 = \sum_{k=1}^K \beta_k(1 - \beta_k) \geq K \cdot \beta_{\min}(1 - \beta_{\min}),$$

$$\beta_{\min} = \min_k \beta_k \beta_k < \frac{1}{2} \beta_{\min} > 0 B_K = \Omega(\sqrt{K}) \beta_{\min} = 0 B_K = \Omega(\sqrt{K}) \quad 0$$

$$\frac{\sum \rho_k}{B_K^3} \leq \frac{1}{B_K} = o\left(\frac{1}{\sqrt{K}}\right).$$

$$\begin{aligned} \text{Lyapunov Berry-Esseen} &O(1/\sqrt{K}) \\ \text{3} &\Theta(1/\sqrt{K}) \end{aligned}$$

$$Z_K := \frac{\sqrt{K}(T_K - \bar{\beta})}{\hat{\sigma}_K},$$

$$\hat{\sigma}_K^2 = \frac{1}{K} \sum_{k=1}^K (Y_k - T_K)^2 Z_K > z_{1-\delta} H_0 z_{1-\delta} 1 - \delta$$

H_0 Berry-Esseen

$$\sup_x |P_{H_0}(Z_K \leq x) - \Phi(x)| \leq \frac{C_0}{B_K} = o\left(\frac{1}{\sqrt{K}}\right).$$

$$\lim_{K \rightarrow \infty} P_{H_0}(Z_K > z_{1-\delta}) = \delta$$

$$H_1 \quad \eta_K = c/\sqrt{K} \quad c > 0 \quad \text{Theorem ??}$$

$$\mathbb{E}[T_K \mid H_1] = \bar{\beta} + \eta_K(\bar{\alpha} - \bar{\beta}) = \bar{\beta} + \frac{c(\bar{\alpha} - \bar{\beta})}{\sqrt{K}}.$$

$$\mathbb{E}[Z_K \mid H_1] = \frac{\sqrt{K} \cdot \eta_K(\bar{\alpha} - \bar{\beta})}{\sigma} + o(1) = \frac{c(\bar{\alpha} - \bar{\beta})}{\sigma} + o(1),$$

$$\sigma^2 = \lim_{K \rightarrow \infty} \mathbb{V}[\sqrt{K} T_K \mid H_0] = \lim_{K \rightarrow \infty} \frac{1}{K} \sum \beta_k(1 - \beta_k)$$

$$P_{H_1}(Z_K > z_{1-\delta}) = 1 - \Phi\left(z_{1-\delta} - \frac{c(\bar{\alpha} - \bar{\beta})}{\sigma}\right) + O\left(\frac{1}{\sqrt{K}}\right).$$

c

$$z_{1-\delta} - \frac{c(\alpha_{\min} - \beta_{\max})}{\sigma} \leq z_{\delta} = -z_{1-\delta},$$

$$c \geq \frac{2\sigma z_{1-\delta}}{\alpha_{\min} - \beta_{\max}} \rightarrow 1 \quad \eta_{\min}(K) = O(1/\sqrt{K})$$

$$4 \quad \Theta(1/\sqrt{K}) \quad \eta = o(1/\sqrt{K}) \quad Z_K \quad \sqrt{K} \cdot \eta(\bar{\alpha} - \bar{\beta}) \rightarrow 0 \text{ Berry-Esseen} \quad Z_K$$

$$H_0 \quad H_1 \quad O(1/\sqrt{K}) \quad 0 \text{ Le Cam} \quad \eta_{\min}(K) = \Omega(1/\sqrt{K})$$

5 Berry-Esseen

$$\liminf_{K \rightarrow \infty} \eta_{\min}(K) \sqrt{K} = \frac{2z_{1-\delta} \sqrt{\beta_{\max}(1 - \beta_{\max})}}{\alpha_{\min} - \beta_{\max}}.$$

$$\delta z_{1-\delta} \sim \sqrt{2 \log(1/\delta)} \quad (??)$$

[Rigorous]

□

:

- **Berry-Esseen** K Berry-Esseen $O(1/\sqrt{K})$ $C_0 \approx 0.5583$ $K \rightarrow \infty$
 $K = 3$ $K = 5$ C_0/B_K 0.3 K $\Theta(1/\sqrt{K})$
- $\hat{\sigma}_K$ Berry-Esseen Slutsky $\hat{\sigma}_K$ Bootstrap
- $\beta_{\min}$ Lyapunov $\beta_{\min} > 0$ $\beta_k = 0$ 0 B_K
- $z_{1-\delta}$ $\hat{\sigma}_K$ $\alpha_{\min}$ $\beta_{\max}$

□

Applications: Theorem ?? provides the **detection sensitivity budget** for fixed-size expert panels. With $K = 3$ experts (a typical crowd-sourcing setup), the minimum detectable noise level is $\eta_{\min} \approx 0.41$ — only very noisy datasets (over 40% label errors) can be reliably detected. With $K = 100$ experts, $\eta_{\min} \approx 0.07$ — still unable to detect noise below 7%. The $\Theta(1/\sqrt{K})$

scaling is a harsh constraint: achieving $\eta_{\min} = 0.01$ requires $K \approx 5 \times 10^4$ experts, far beyond practical budgets. This theorem fundamentally limits *pure* expert-based noise detection and motivates hybrid approaches: combine a small expert panel ($K \approx 20$ – 50 , detecting $\eta \approx 0.10$ – 0.15) with Cercis-based statistical detection (which can flag samples for expert review at lower effective η via T5’s optimal sampling). In practice, the theorem guides the “expert escalation threshold”: set the automated detector’s sensitivity to flag samples at the noise level $\eta_{\min}(K)$ that the available expert panel can verify, ensuring the human-in-the-loop component is never asked to adjudicate below its detection limit.

Remark 3.5. *The $\Theta(1/\sqrt{K})$ scaling has immediate practical implications:*

- *To halve the minimum detectable noise level, you need four times as many experts;*
- *To detect $\eta = 0.01$ (1% noise) with $\alpha = 0.8, \beta = 0.2$, you need $K \approx 2 \log(1/\delta)/(0.01^2 \cdot 0.36) \approx 5 \times 10^4$ experts — an impossibility in most domains, motivating the need for complementary detection strategies.*

[Rigorous]

3.3 Correlated Experts and Audit Diversity

Theorem 3.6 (Audit Diversity — Effective Expert Count — Formal Statement). *Relax Assumption ?? and allow pairwise Pearson correlation $\rho_{ij} = \text{Corr}(Y_i, Y_j)$ between expert judgments Y_i, Y_j . Assume equal marginal variances $\mathbb{V}[Y_k] = \sigma_0^2$ for all k . Then:*

(i) *The variance of the majority statistic is*

$$\mathbb{V}[T_K] = \frac{\sigma_0^2}{K} [1 + (K-1)\bar{\rho}], \quad (12)$$

where $\bar{\rho} = \frac{2}{K(K-1)} \sum_{i < j} \rho_{ij}$ is the mean pairwise correlation.

(ii) *The effective independent expert count is*

$$K_{\text{eff}} = \frac{K}{1 + (K-1)\bar{\rho}}. \quad (13)$$

(iii) *The minimum detectable noise level becomes*

$$\eta_{\min}(K, \bar{\rho}) \geq \sqrt{\frac{2 \log(1/\delta)}{K_{\text{eff}} (\alpha_{\min} - \beta_{\max})^2}}. \quad (14)$$

Proof.

(P1) Pearson $\rho_{ij} = \text{Corr}(Y_i, Y_j)$ $Y_k = 1_{\{J_k = \text{noise}\}}$

(P2) $\mathbb{V}[Y_k] = \sigma_0^2 > 0$ k

(P3)

(P4) Assumptions $\alpha_k \geq \alpha_{\min}$ $\beta_k \leq \beta_{\max}$

$$1 \quad T_K = \frac{1}{K} \sum_{k=1}^K Y_k$$

$$\mathbb{V}[T_K] = \frac{1}{K^2} \left[\sum_{k=1}^K \mathbb{V}[Y_k] + \sum_{i \neq j} \text{Cov}(Y_i, Y_j) \right].$$

$$\text{P2} \quad \text{Cov}(Y_i, Y_j) = \rho_{ij} \sigma_0^2$$

$$\begin{aligned} \mathbb{V}[T_K] &= \frac{1}{K^2} \left[K \sigma_0^2 + \sigma_0^2 \sum_{i \neq j} \rho_{ij} \right] \\ &= \frac{\sigma_0^2}{K^2} \left[K + \sum_{i \neq j} \rho_{ij} \right]. \end{aligned}$$

$$\sum_{i \neq j} \rho_{ij} = K(K-1)\bar{\rho}$$

$$\mathbb{V}[T_K] = \frac{\sigma_0^2}{K^2} [K + K(K-1)\bar{\rho}] = \frac{\sigma_0^2}{K} [1 + (K-1)\bar{\rho}].$$

(??)

$$2 \quad \rho_{ij} = 0 \quad \mathbb{V}[T_K]_{\text{ind}} = \sigma_0^2 / K \quad [1 + (K-1)\bar{\rho}] \quad K_{\text{eff}} \quad \mathbb{V}[T_K] = \sigma_0^2 / K_{\text{eff}}$$

$$\frac{\sigma_0^2}{K_{\text{eff}}} = \frac{\sigma_0^2}{K} [1 + (K-1)\bar{\rho}] \implies K_{\text{eff}} = \frac{K}{1 + (K-1)\bar{\rho}}.$$

(??)

$$3 \quad \text{Theorem } ?? \quad K \quad K_{\text{eff}} \quad K_{\min}$$

$$\eta_{\min}(K, \bar{\rho}) \geq \sqrt{\frac{2 \log(1/\delta)}{K_{\text{eff}} (\alpha_{\min} - \beta_{\max})^2}}.$$

(??)

$$4 \text{Jensen} \quad \sigma_k^2 = \mathbb{V}[Y_k]$$

$$\mathbb{V}[T_K] = \frac{1}{K^2} \left[\sum \sigma_k^2 + \sum_{i \neq j} \rho_{ij} \sigma_i \sigma_j \right].$$

$$\bar{\sigma}^2 = \frac{1}{K} \sum \sigma_k^2 \quad \bar{\rho} = \frac{2}{K(K-1)} \sum_{i < j} \rho_{ij} \quad \text{Cauchy-Schwarz Jensen}$$

$$\sum_{i \neq j} \rho_{ij} \sigma_i \sigma_j \leq \sum_{i \neq j} |\rho_{ij}| \sigma_i \sigma_j \leq \max_{i \neq j} |\rho_{ij}| \sum_{i \neq j} \sigma_i \sigma_j.$$

$$\max |\rho_{ij}| \leq \bar{\rho} \sum_{i \neq j} \sigma_i \sigma_j \leq K(K-1) \bar{\sigma}^2 \quad \text{AM-GM}$$

$$\mathbb{V}[T_K] \leq \frac{\bar{\sigma}^2}{K} [1 + (K-1)\bar{\rho}].$$

$$\bar{\rho} \quad \bar{\sigma}^2 \quad K_{\text{eff}}$$

[Conditionally Rigorous]

□

:

- $K_{\text{eff}} = K/[1 + (K-1)\bar{\rho}] \quad \rho_{ij} = \rho \quad \mathbb{V}[Y_k] = \sigma_0^2 \quad \rho_{ij} \quad \bar{\rho} \quad \rho = 1 \quad \rho = 0 \quad \bar{\rho} \approx \frac{1}{2}$
- $\bar{\rho} \quad K_{\text{eff}} \quad K_{\text{eff}}$
- $\varepsilon \quad \text{Assumption ??} \quad \varepsilon \quad \rho_{ij} = \rho_{ij}^{(\varepsilon)} + \rho_{ij}^{(\text{shared})}$
- $\sigma_i^2 \neq \sigma_j^2 \quad (??)$

$$\mathbb{V}[T_K] = \frac{1}{K^2} \left[\sum \sigma_k^2 + \sum_{i \neq j} \rho_{ij} \sigma_i \sigma_j \right],$$

$$K_{\text{eff}}$$

□

Applications: Theorem ?? provides the **diversity dividend** for expert panel composition. The effective expert count $K_{\text{eff}} = K/[1 + (K-1)\bar{\rho}]$ reveals that correlated experts provide rapidly diminishing returns: with $\bar{\rho} = 0.3$, a panel of $K = 20$ experts is equivalent to only $K_{\text{eff}} \approx 3.3$ independent experts. This has immediate operational implications: (i) recruit experts from maximally diverse backgrounds (different training institutions, different specialties, different geographical regions) to minimize $\bar{\rho}$; (ii) use different AI audit modules with non-overlapping training data and distinct architectures to achieve low inter-model correlation; (iii) regularly measure $\bar{\rho}$ on calibration sets and rotate out highly correlated experts. In practice, a panel of $K = 10$ experts with $\bar{\rho} = 0.1$ ($K_{\text{eff}} \approx 5.3$) outperforms a panel of $K = 50$ experts with $\bar{\rho} = 0.5$ ($K_{\text{eff}} \approx 2.0$). The theorem also provides a procurement metric: when evaluating expert panel providers, compute K_{eff} from their correlation matrix rather than comparing raw K — effective diversity, not headcount, determines audit power.

Open Problem 3.7 (Optimal Expert Selection). *Given a pool of N candidate experts with known pairwise correlations ρ_{ij} and individual quality parameters (α_k, β_k) , select a subset of size K that maximizes K_{eff} . This is conjectured to be a submodular maximization problem with cardinality constraint, analogous to sensor placement [?]. The exact complexity class and approximation ratio are **open**. [Open]*

3.4 Model Complexity–Audit Cost Duality

Conjecture 3.8 (Model Complexity–Audit Cost Duality). *Let M be the model’s audit complexity (Definition ??). There exists a universal constant $c > 0$ such that for any auditor with expert budget K and any model with complexity M ,*

$$K_{\min} \cdot \eta_{\min}^2 \geq c \cdot \frac{M}{\log M} \cdot \log(1/\delta). \quad (15)$$

*Equivalently: **you cannot simultaneously have low audit cost, detect small noise, and audit a complex model** — these three quantities satisfy a fundamental uncertainty relation.*

Remark 3.9. *The conjectured form draws on statistical learning theory’s standard duality between sample complexity and model complexity (VC dimension). Here, K replaces sample size n , and η^2 replaces excess risk ε . The $M/\log M$ factor is characteristic of the minimax rate for aggregate inference in high-dimensional models [?]. If the conjecture is true, it provides an **economic impossibility theorem** for large-scale auditing: beyond a certain model complexity, reliable noise detection requires more experts than exist. [Open]*

4 Discussion

4.1 The Expert Supply Constraint

The $\Theta(1/\sqrt{K})$ scaling exposes a harsh reality: detecting small noise levels ($\eta < 0.05$) with high confidence requires hundreds to thousands of independent experts. For most practical domains, this is infeasible. Three mitigation strategies emerge from the theory:

- (i) **Improve expert quality** ($\alpha_{\min} \uparrow, \beta_{\max} \downarrow$): the bound depends on $(\alpha_{\min} - \beta_{\max})^{-2}$, so small improvements in expert quality yield large reductions in $K_{\min}$;

- (ii) **Reduce expert correlation** ($\bar{\rho} \downarrow$): diversifying expert backgrounds, training data, and architectures increases K_{eff} without increasing K ;
- (iii) **Combine with Cercis-based detection**: Theorem ?? applies to *pure* expert-based detection; the Cercis Score provides a complementary signal that can reduce the required K .

4.2 Connection to Other SCX Theorems

- **T5 (Active Learning)**: T5 provides sample complexity $n_{\min}$; AC-Theorem provides expert complexity $K_{\min}$. Together, $(n_{\min}, K_{\min})$ fully specify the resource requirements of an audit.
- **FA-Theorem (Federated Audit)**: AC-Theorem (iii)’s K_{eff} quantifies the diversity benefit of federated expert pools.
- **HC-Theorem (Human–AI)**: Human experts have measurable (α, β) parameters, making AC-Theorem directly applicable to budget allocation in human–AI audit protocols.
- **AE-Theorem (Audit Entropy)**: The Landauer energy cost times K experts gives the total thermodynamic cost of an audit — $E_{\text{total}} \geq K \cdot k_B T \cdot \Delta H \cdot \ln 2$.

5 Conclusion

AC-Theorem quantifies the expert-resource requirements of SCX auditing, establishing a $\Theta(1/\sqrt{K})$ noise-detection threshold, an effective expert count under correlation, and a conjectured duality between model complexity and audit cost. The rigorous bounds provide operational guidance for audit deployment; the open problems — optimal expert selection and the model-complexity duality — define the frontier of SCX resource theory.

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